III. GRAPHICAL REPRESENTATIONS :

→ HISTOGRAM
→ BOXPLOTS
→ SCATTER PLOTS

1. HISTOGRAM : USEFUL FOR IDENTIFYING PATTENS, OUTLIERS & SKEWNESS.

- Histograms are a type of bar chart that represents the frequency distribution of a dataset.
- They display the distribution of data by showing the number of data points that fall within a specified range of values, known as **BINS**.

- Each bar in a histogram represents the frequency of data within that bin.

* In Histograms "**pattern**" refers to overall shape of the data distribution.

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* Skewness refers to the asymmetry in the distribution of data points; If the tail on one side is longer or flatter, the distribution is said to be skewed.
POSITIVE (+ve) skew has a long right tail,
NEGATIVE (-ve) skew has a long left tail.

* **OUTLIER**

→ An outlier is a value that sits away from most of other values in a dataset and shows up as a separate bar or empty gap in a histogram.
→ Outliers may be true extreme values/cases, mistakes in recording the data or signs that a different process produced those values.

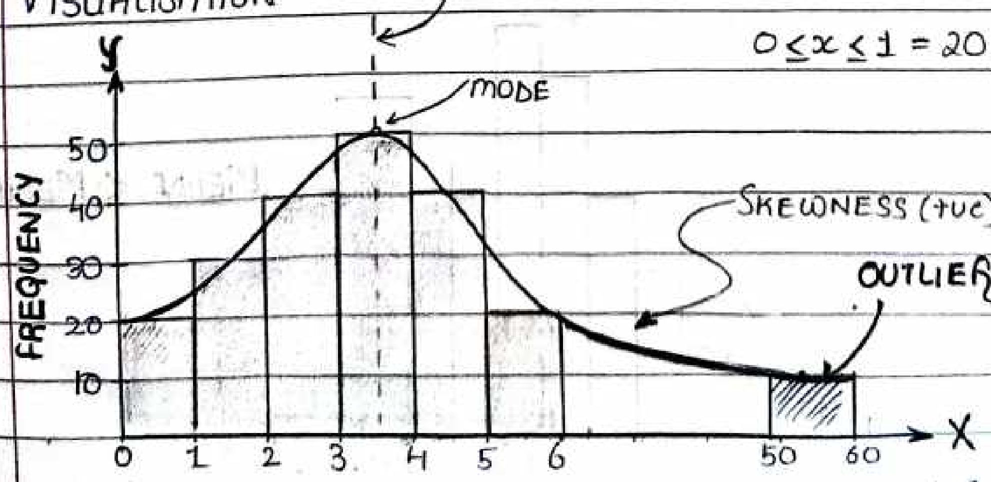
Eg:- • If most people took 5-15 min on a test and one person 60 minutes, 60 is an OUTLINER.

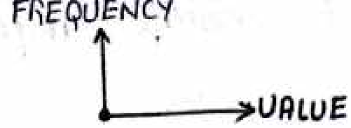
• If daily sales are usually 50-100 units and one day shows 1000, that day is an outlier.

→ Outliers can change averages and give a misleading picture of data.
They can be mistakes, rare events or sign of different process.

* VISUALISATION

• Dataset :- 0.1, 0.2, 0.3, 0.11, 0.22, ...
$0 \leq x \leq 1$ = 20 values b/w

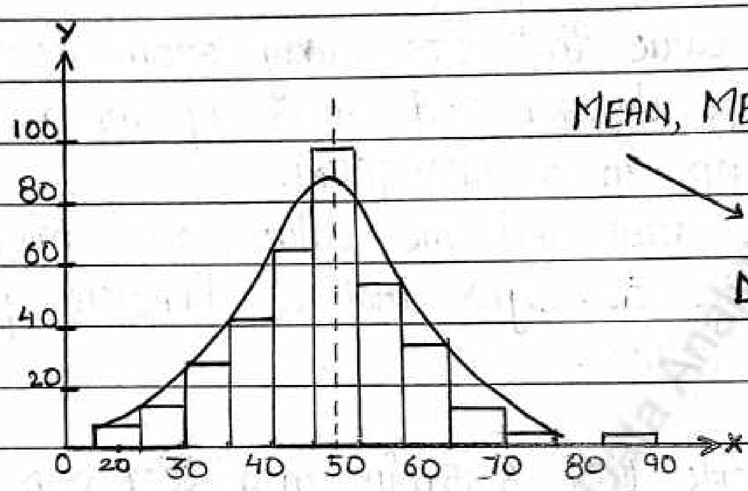




* TYPES OF HISTOGRAMS :-

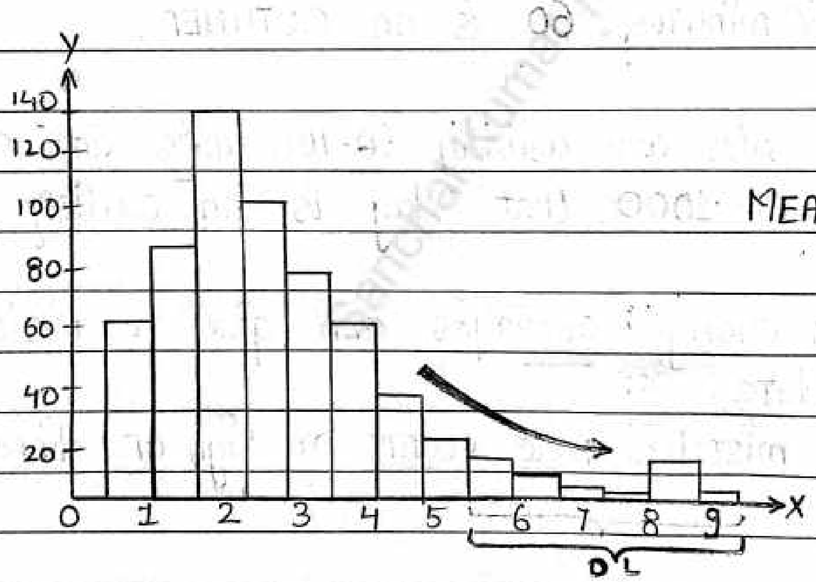
- SYMMETRIC HISTOGRAM ≡ NORMAL DISTRIBUTION
- RIGHT SKEWED HISTOGRAM ≡ +VE SKEWED
- LEFT SKEWED HISTOGRAM ≡ −VE SKEWED

1.



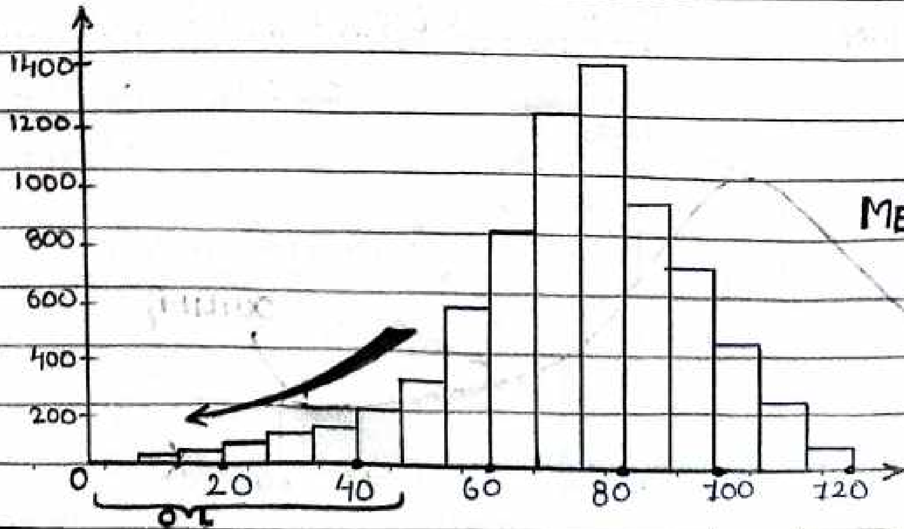
MEAN, MEDIAN & MODE → CENTRE OF DISTRIBUTION

2.



MEAN > MEDIAN

3.



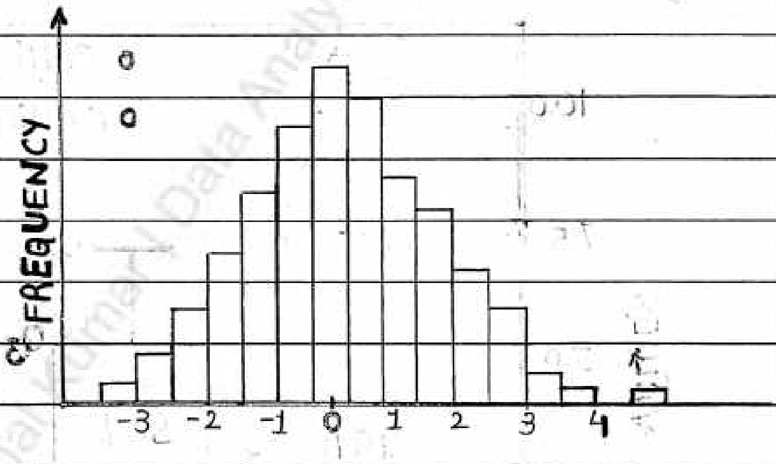
MEAN < MEDIAN

2. R.S.H. where data points towards the higher end, around 6-9 have comparitively lower frequency.
These higher values despite their lower frequency, can skew (the distribution [making distribution unsymmetrical / biased), making them potential outliers.

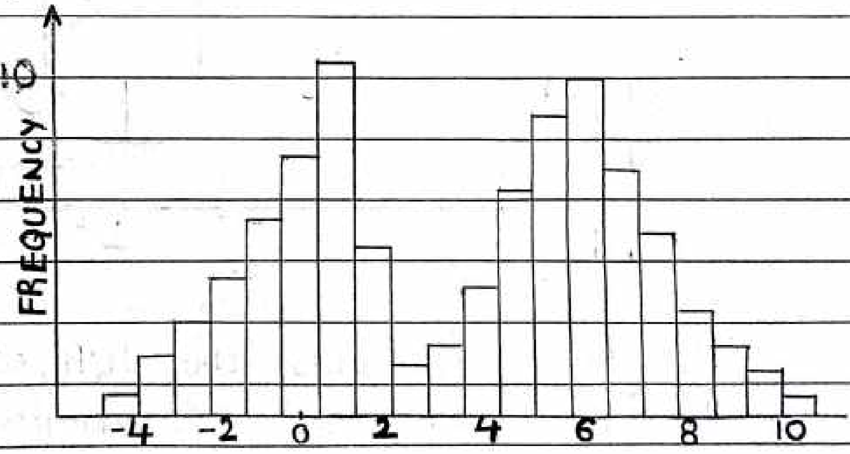
3. L.S.H. you usually see more data points concentrated on the right. The tail on the left extends further.
Data values with lower frequency towards the left side can be the potential outliers.

* Types of Histograms based on Number of Modes :-

1. Unimodal ✓
1 single peak.

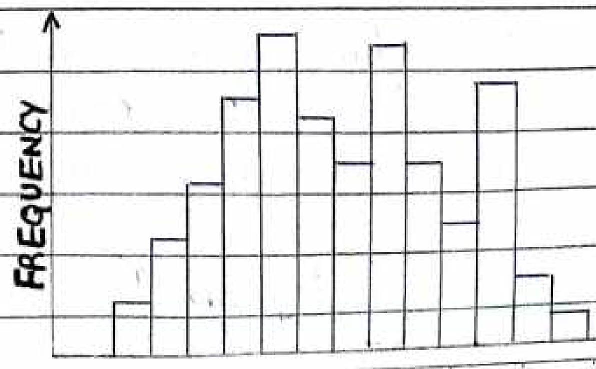


2. Bimodal Histogram
2 peaks



3. Multimodal Histogram
Multiple Peaks.

1:17:31



* UPPER WHISKER = (Q3 + 1.5 IQR)
* LOWER WHISKER = (Q_1 − 1.5 IQR)

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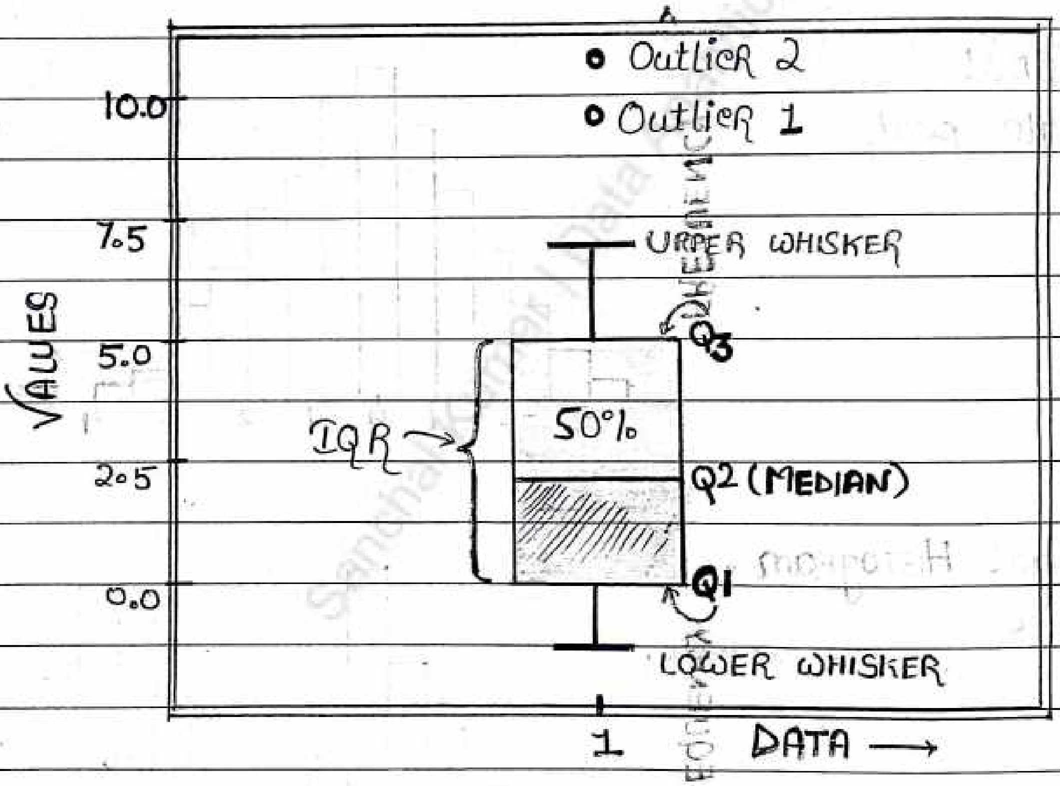
2. BOXPLOT

- Boxplots are great for showing data spread and identifying outliers
- Each Boxplot has → QUARTILES,
  → WHISKERS,
  → POTENTIAL OUTLIERS usually shown as individual points.



→ The box represents the IQR, which is range between the first and the 3rd quartiles (Q1 & Q3)

→ The line inside the box indicates the median.
→ The "whiskers" extend to the smallest and largest values within 1.5 times the IQR from the quartiles.

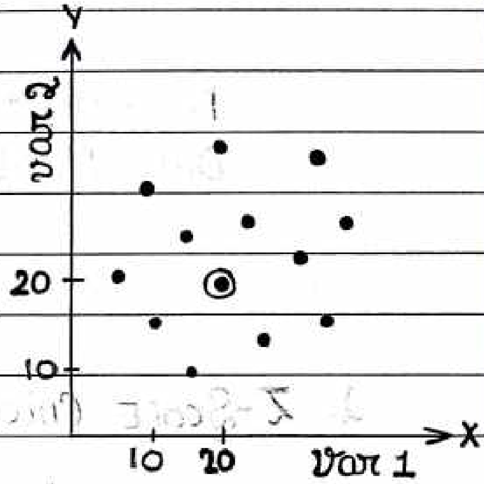
→ Values beyond the whisker range are potential outliers.

The data b/w the Q3 and Upper Whisker represents part of the upper 25% of the dataset but not necessarily all of it, since extreme values or outliers could extend further.

3. Scatter Plot

→ A scatter plot visualizes the relationship between two continuous variables.

→ Each point → observation, with its position determined by the values of those variables.

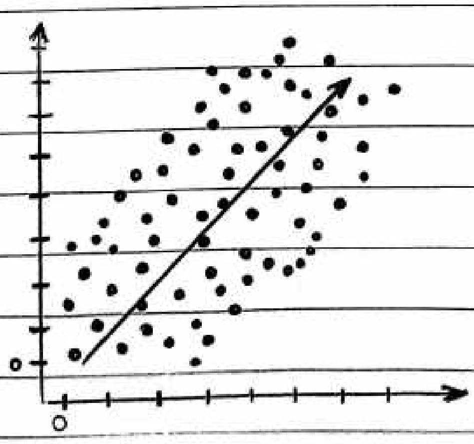


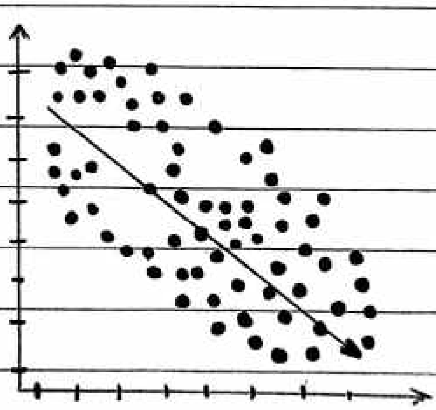
• var 1 - 20
• var 2 - 20

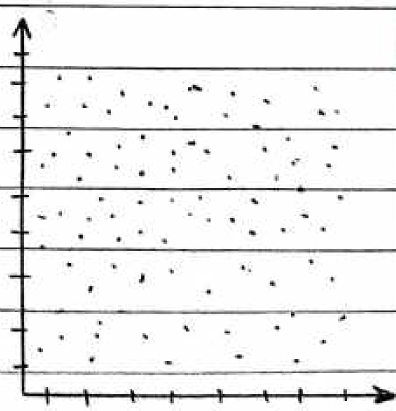
→ By examining the pattern of points we can identify :-

- Trends
- Correlations or clusters &
- Spot any outliers.

* Relationship Between Variables :-







Positive Correlation

• var 1 ↑ var 2 ↑

Negative Correlation

var 1 ↑ var 2 ↓

No Correlation

var 1 X var 2